Automated Parcel Damage Detection System for E-Commerce Fulfillment Centres

Executive Summary

Overview

This project presents an AI-powered damage detection and classification pipeline for automated parcel inspection. The system combines a custom-trained YOLOv8 model, Roboflow serverless inference platform, Python-based processing pipeline, and Power BI dashboard to identify and classify parcel damage efficiently. The pipeline processes input images, predicts damage severity, stores results in structured datasets, and provides actionable insights through analytics and reporting.

Key Capabilities

Automated batch image processing.

YOLOv8-based damage detection.

Severity classification (Severity_1 & Severity_2).

Structured CSV output logging.

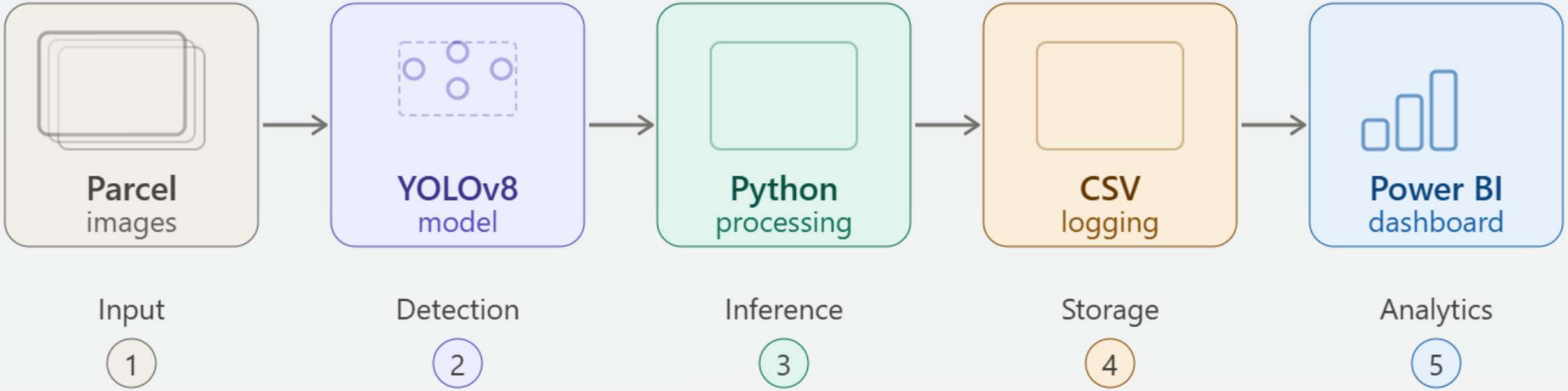
Live image simulation for real-time processing.

Data validation before dashboard integration.

Scalable and modular system architecture.

AI damage detection pipeline

Click any block to learn more



System Architecture

Overview

The AI-powered damage detection system follows a modular architecture that processes parcel images through multiple stages. The system detects damage, classifies severity levels, stores results in a structured format, and provides analytics through Power BI dashboards.

Key Points

Automated parcel inspection using AI.
YOLOv8 model for damage detection.
Python-based inference engine.
Structured CSV data logging.
Data validation before analytics.
Power BI dashboard integration.

Component	Technology
Dataset	Roboflow
Detection Model	YOLOv8
Inference Engine	Python
Data Storage	CSV
Analytics	Power BI

Dataset & Model Configuration

Overview

The parcel damage detection model was trained using a custom Roboflow dataset containing annotated images of damaged parcels. Each image was labeled with severity classes to help the model learn different levels of package damage. The trained YOLOv8 model uses these annotations to accurately identify and classify damage during inference.

Key Points

Custom dataset created using Roboflow.

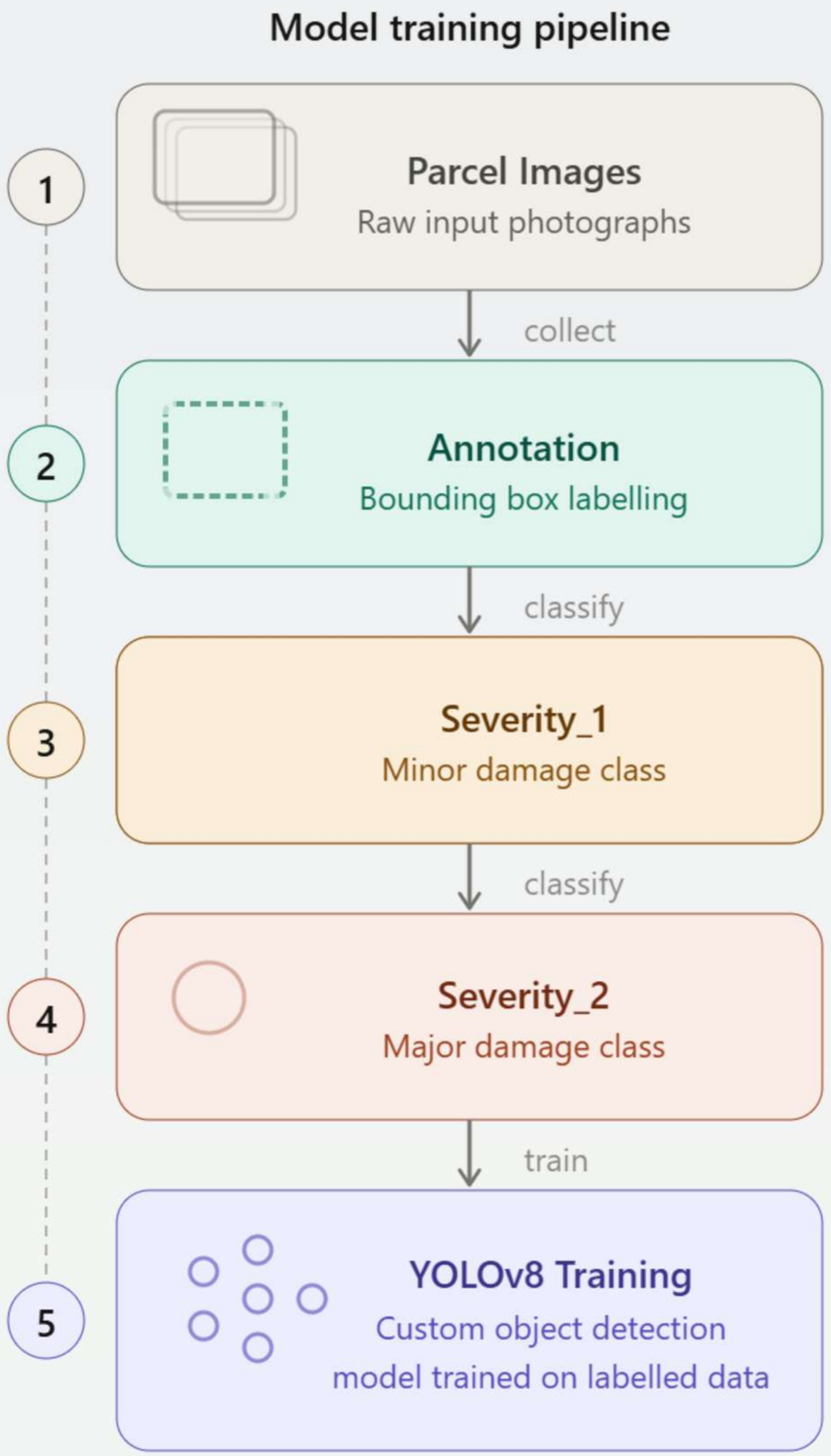
Images annotated with bounding boxes.

Two severity classes used for training.

YOLOv8 model trained for damage detection.

Supports confidence-based predictions.

Enables automatic damage classification.


Model training pipeline
1
Parcel Images
Raw input photographs
collect
2
Annotation
Bounding box labelling
classify
3
Severity_1
Minor damage class
classify
4
Severity_2
Major damage class
train
5
YOLOv8 Training
Custom object detection
model trained on labelled data

Damage Classes

Class	Description
Severity_1	Minor surface damage
Severity_2	Major structural damage

Model Features

Feature	Description
Detection	Identifies parcel damage
Classification	Severity-based prediction
Confidence Score	Prediction certainty
Bounding Box	Damage location detection

Python Inference Setup

Overview

The Python inference engine was developed to connect the trained YOLOv8 model with the Roboflow Serverless API. This module processes parcel images, sends them to the model for prediction, and retrieves damage classification results in real time. It serves as the core component of the automated damage detection pipeline.

Key Points

Installed Roboflow Python SDK.

Configured API key and workflow ID.

Connected to YOLOv8 model.

Processed images automatically.

Generated class labels and confidence scores.

Supported Severity_1 and Severity_2 detection.

Implementation Details

Component	Purpose
Roboflow SDK	Model connectivity
Roboflow SDK	Model connectivity
Roboflow SDK	Model connectivity
Roboflow SDK	Model connectivity
Roboflow SDK	Model connectivity

Results

Successfully processed all test images.

Detected Severity_1 and Severity_2 classes.

Confidence scores ranged from 0.40 to 0.88.

Prediction results generated successfully.

Structured Output Logging

Overview

The structured output logging module captures all prediction results generated by the YOLOv8 model and stores them in a CSV file. This ensures that every detection is recorded with complete metadata, making the data suitable for analytics, reporting, and Power BI dashboard integration.

Key Points

Stores prediction results in CSV format.

Records image name and class label.

Logs confidence score for each detection.

Captures bounding box coordinates.

Maintains timestamps for traceability.

Supports multiple detections per image.

Logged Fields

Field Name	Description
image_name	Input image filename
class_label	Predicted damage class
confidence	Model confidence score
x, y	Bounding box coordinates
width, height	Bounding box dimensions
timestamp	Detection date and time

Results

Generated structured CSV output.

Logged all prediction details successfully.

Supported multi-detection images.

Data prepared for Power BI analytics.

Outcome

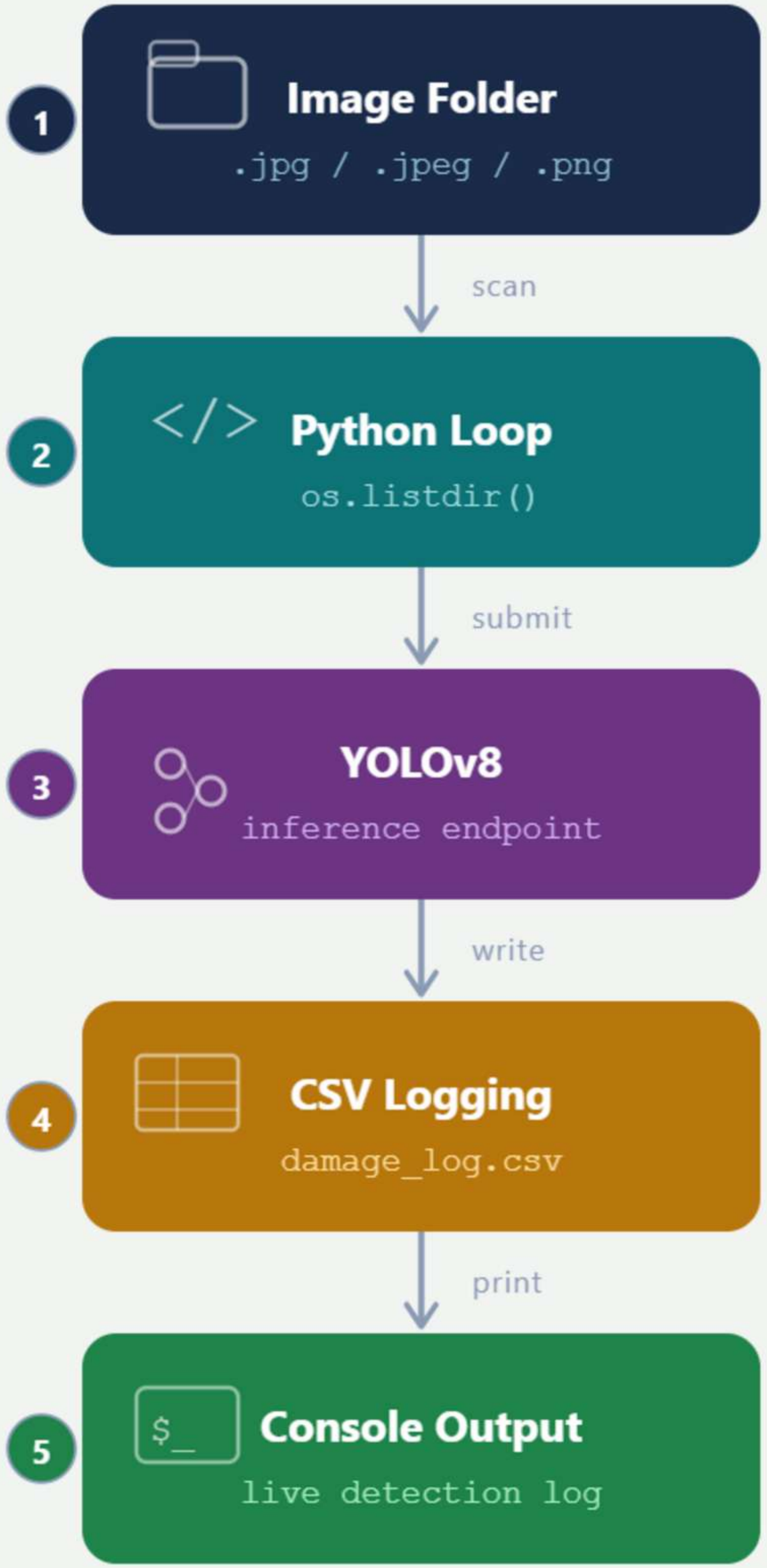
✔ Structured output generation

✔ Complete prediction logging

✔ CSV dataset creation

✔ Analytics-ready data storage

Workflow Diagram



Live Image Simulation & Batch Pipeline

Overview

The live image simulation module was developed to mimic a real-time camera feed using a folder of parcel images. Images are processed sequentially through the YOLOv8 model, and prediction results are logged immediately. This simulates how the system would operate in an actual fulfillment center environment.

Key Points

Simulates real-time image processing.

Reads images from input folder automatically.

Supports JPG, JPEG, and PNG formats.

Processes images sequentially.

Updates CSV log after each detection.

Displays live results on console.

Includes error handling for uninterrupted execution.

Live Image Simulation Processing

Average Precision by Class (mAP50)

These metrics are reported by the model provider and may not follow industry-standard evaluation techniques. Learn more about model evaluation best practices and important caveats here.

Validation Set | Test Set

all	73.0%
severity_1	71.0%
severity_2	75.0%

Average Precision by Class (mAP50)

These metrics are reported by the model provider and may not follow industry-standard evaluation techniques. Learn more about model evaluation best practices and important caveats here.

Validation Set | **Test Set**

all	69.0%
severity_1	71.0%
severity_2	67.0%

Pipeline Validation & End-to-End Integration Testing

Overview

The final stage of the project focused on validating the complete pipeline to ensure data accuracy, consistency, and readiness for Power BI integration. Multiple validation checks were performed on the generated CSV dataset to confirm that all prediction records were correctly stored and formatted.

Key Points

Verified all processed images were logged.

Checked for missing or null values.

Validated confidence score ranges.

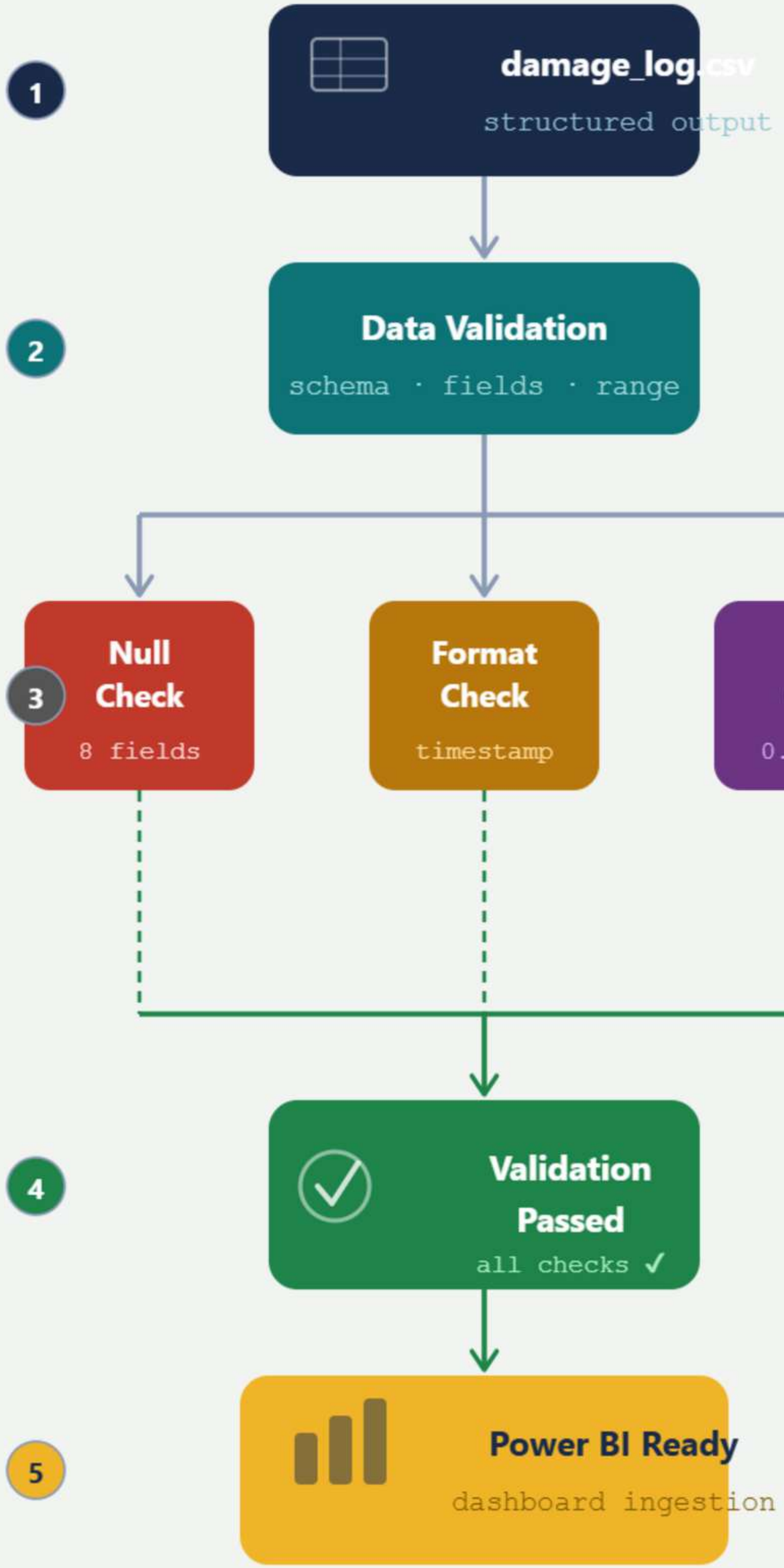
Confirmed timestamp consistency.

Verified multi-detection image records.

Reviewed final CSV structure in Excel.

Uploaded project files to GitHub.

Validation Flow Diagram



Power BI Dashboard & Analytics

Overview

Power BI was integrated to visualize parcel damage detection results and provide actionable insights. The dashboard enables users to monitor damage trends, severity distribution, detection frequency, and overall system performance through interactive charts and reports.

Key Points

Connected CSV output to Power BI.

Visualized damage detection statistics.

Monitored severity distribution.

Displayed image processing performance.

Generated interactive reports.

Supported data-driven decision making.

Graph for this Slide

A Pie Chart works best:

Severity_1 → 43%

Severity_2 → 57%

Results & Performance Analysis

Overview

The developed parcel damage detection system successfully identified and classified damaged parcels using the trained YOLOv8 model. The system achieved reliable detection performance during testing and demonstrated the capability to automate parcel inspection and damage assessment.

Key Points

Successfully detected parcel damages.

Classified damages into severity levels.

Generated structured prediction outputs.

Achieved consistent inference performance.

Supported real-time simulation.

Enabled automated inspection workflow.

THANK YOU